

The Solver Tolerance Is Not a Solver Parameter:

Separating Numerical Noise from Thermodynamic Inconsistency
in Cycle-Consistency Tests on Metabolic Circuits

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Abstract

A metabolic network at steady state can be written as a resistive circuit, with chemical potentials as node voltages and reaction fluxes as branch currents. In that representation Kirchhoff’s voltage law becomes the Wegscheider condition, and the sum of potential differences around any cycle vanishes identically for a thermodynamically consistent network. This makes cycle sums an attractive diagnostic: a nonzero cycle sum witnesses inconsistency, and *where* it is nonzero localises the inconsistency to a region of the network. Any such test needs a tolerance ε below which a cycle sum counts as zero. It is normal practice to fix ε at a round number — 10^{-6} is typical — and to treat it as a property of the solver.

We show this is a category error with a measurable cost, and we replace it. Our first result is negative and structural: the cycle sum of a consistent circuit is *identically* zero as an algebraic identity, so its computed value is pure numerical noise, and the natural scale of that noise is not a constant but a quantity that grows with the network’s potential range and cycle length (Theorem 3.2). We give the bound $\varepsilon_{\text{num}}(\ell) \leq \gamma_\ell \Lambda$ with $\gamma_\ell = \ell u / (1 - \ell u)$ and Λ the largest absolute potential on the cycle, and verify it is attained to within a small factor. A fixed ε is therefore simultaneously too tight on long high-potential cycles, where it manufactures false positives, and too loose on short low-potential ones, where it hides real defects. On a synthetic network spanning realistic ranges, a fixed $\varepsilon = 10^{-6}$ yields a false-positive rate of 0.41 on the longest cycles while missing 28% of genuine perturbations on the shortest.

Our second result is positive. We define the *cycle-local tolerance* $\varepsilon^*(\ell) = \max\{\varepsilon_{\text{num}}(\ell), \varepsilon_{\text{data}}(\ell)\}$, where $\varepsilon_{\text{data}}$ propagates the reported uncertainty of the underlying thermodynamic data through the same cycle sum, and prove that a test thresholded at ε^* has a numerical false-positive rate bounded by the arithmetic alone (Theorem 6.2) and detects every perturbation exceeding $2\varepsilon^*$ (Theorem 6.3). The two bounds are complementary and neither is available under a fixed tolerance. We show that the resulting classification is three-valued and that the third value is not optional: a cycle whose sum lies between the numerical and data floors is *undecidable from the data supplied*, and reporting it as either consistent or inconsistent is unwarranted (Theorem 6.6).

Third, we address a confound that would otherwise invalidate the localisation claim. Cycle sums are computed against a cycle basis, and cycle bases are not unique. We prove that the *verdict* is basis-independent — a network is consistent or not, whatever basis is chosen (Theorem 7.1) — but that the *localisation* is not: the set of flagged cycles depends on the basis, and a defect can be made to appear in one cycle or spread across several by choosing differently (Theorem 7.2). We give the invariant that survives: the set of edges lying on every flagged cycle of some minimum cycle basis, which we call the *witness set*, and show it contains the perturbed edge whenever detection succeeds.

We specify the resulting protocol to executable detail, give it as a program in a small typed language whose type system makes an intolerance-annotated verdict unwritable, and state quantitative success and failure criteria in advance. We distinguish, as the design requires, between an experiment that returns a negative result and one that is *invalid*: a healthy circuit failing its own consistency check indicates the data, not the theory, and we give the check that separates these before any diagnostic claim is evaluated.

Keywords: Wegscheider condition; Kirchhoff’s laws; metabolic networks; cycle basis; floating-point error analysis; detection limit; tolerance selection; three-valued classification; domain-specific languages.

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1 Introduction

1.1 A test with a free parameter nobody chooses

Consider a biochemical reaction network at steady state. Assign to each species i a chemical potential μ_i and to each reaction (i, j) a flux J_{ij} . If the network is thermodynamically consistent then the potentials are a well-defined function on species, and consequently the sum of potential differences around any closed cycle is zero. This is Kirchhoff’s voltage law in the circuit reading of the network (Oster et al., 1971; Peusner, 1986), and in the chemical reading it is the Wegscheider condition on the rate constants (Schuster and Schuster, 1989; Wegscheider, 1901).

The identity suggests a diagnostic. Compute, for each cycle ℓ in the network, the cycle sum

$$H_\ell = \sum_{(i,j) \in \ell} (\mu_j - \mu_i), \quad (1)$$

and declare the network inconsistent when some $|H_\ell|$ is “too large”. Because the sum is attached to a specific cycle, a positive result carries a location: it says not merely that something is wrong but that something is wrong *here*, in this loop. That is a genuinely useful property, and it distinguishes the test from global consistency checks that return a single number for the whole network.

Everything then turns on “too large”. The test requires a tolerance ε , and the near-universal practice is to fix one — 10^{-6} , 10^{-9} , machine epsilon times some factor — and to describe it as a solver setting. This paper argues that treating ε as a solver setting is a category error, that the error has a measurable cost in both directions, and that the correct ε is not a constant at all.

1.2 Why a fixed tolerance cannot be right

The argument is short enough to give here in full.

For a consistent network, (1) is not approximately zero; it is *identically* zero, because it telescopes. Writing the cycle as $i_1 \rightarrow i_2 \rightarrow \cdots \rightarrow i_m \rightarrow i_1$,

$$\sum_{r=1}^m (\mu_{i_{r+1}} - \mu_{i_r}) = \mu_{i_1} - \mu_{i_1} = 0, \quad i_{m+1} = i_1,$$

identically in the values μ , with no hypothesis about them beyond their being a function on vertices. Whatever a floating-point computation returns for this quantity is therefore *entirely* rounding error. It contains no information about the network.

But the magnitude of that rounding error is not a constant. Summing m terms each formed by subtracting numbers of magnitude up to Λ accumulates absolute error that grows with both m and Λ (Goldberg, 1991; Higham, 2002). A cycle of length 20 through species whose potentials span ± 2000 kJ/mol will return a computed sum orders of magnitude larger than a triangle through species spanning ± 10 , and both are exactly zero as mathematics. A single threshold applied to both must be wrong for at least one of them. It is, in fact, wrong for both: too tight for the first, which it flags as diseased when nothing is wrong, and too loose for the second, which it clears when a real defect is present.

This is not a subtle effect at the margins. In Section 5 we measure it: at $\varepsilon = 10^{-6}$ on networks with realistic potential ranges, long cycles are flagged spuriously at rate 0.41 while short cycles miss 28% of injected defects. The test does not degrade gracefully; it fails in opposite directions on different parts of the same network, which is the worst possible failure mode for a diagnostic that claims to localise.

1.3 What replaces it

If the tolerance cannot be a constant, it must be computed. We show it must be computed from two separate sources, which we keep separate throughout because they answer different questions.

The first is *numerical*: given the arithmetic and the magnitudes involved, how large can the computed cycle sum be when the true one is zero? This is a standard forward error analysis (Higham, 2002) and we carry it out in Section 3, obtaining a bound $\varepsilon_{\text{num}}(\ell)$ that depends on the cycle’s length and potential range.

The second is *empirical*: the potentials are not exact. They are computed from standard free energies and concentrations, both of which carry reported uncertainty (Alberty, 1998; Flamholz et al., 2012; Noor et al., 2013). A cycle sum smaller than the uncertainty propagated through it is not distinguishable from zero *given the data*, regardless of how exactly the arithmetic was performed. This gives a second floor $\varepsilon_{\text{data}}(\ell)$, developed in Section 4, and it is typically the larger of the two by several orders of magnitude.

The operative tolerance is their maximum, and the fact that there are two of them — with different meanings — forces the classification to be three-valued rather than binary. A cycle sum below both floors is consistent. Above both, inconsistent. Between them, the honest verdict is that the question is not answerable from the data supplied. We prove in Theorem 6.6 that this third class is non-empty on realistic inputs, which is what makes it a substantive part of the test rather than a formality.

1.4 The localisation confound

A cycle-consistency test on a network with $|E|$ edges and $|V|$ vertices has $|E| - |V| + 1$ independent cycles, and a choice of which ones (Diestel, 2017; Kavitha et al., 2009). The choice is not canonical. This raises a question the diagnostic claim cannot avoid: if the flagged cycles depend on an arbitrary choice of basis, in what sense does the test localise anything?

We answer this in Section 7. The verdict is safe: whether *any* cycle is inconsistent is a property of the network, not of the basis (Theorem 7.1), because the cycle space is spanned by any basis and a nonzero functional cannot vanish on a spanning set. The localisation is not safe: we construct networks where one basis attributes a defect to a single cycle and another spreads it over three (Theorem 7.2). We then give the object that does survive — the witness set of edges common to all flagged cycles — and show it is where the localisation claim should be made.

1.5 Contributions

1. **The noise scale is derived, not assumed** (Theorem 3.2): for a consistent circuit the computed cycle sum obeys $|\hat{H}_\ell| \leq \gamma_\ell \Lambda_\ell$ with $\gamma_\ell = \ell u / (1 - \ell u)$, and this is attained to within a factor we measure.
2. **A fixed tolerance fails in both directions simultaneously** (Section 5), with the rates quantified on networks of realistic range.
3. **The cycle-local tolerance $\varepsilon^*(\ell)$** , with a false-positive bound (Theorem 6.2) and a detection guarantee (Theorem 6.3) that a fixed tolerance cannot both have.
4. **The trichotomy is forced** (Theorem 6.6): two floors of different meaning make a two-valued verdict unsound, and the undecidable class is non-empty.
5. **Verdict is basis-independent; localisation is not** (Theorem 7.1, Theorem 7.2), with the witness set as the invariant that survives (Theorem 7.4, Theorem 7.5).
6. **An executable protocol** with pre-registered success and failure criteria (Section 10), expressed in a typed language whose type system makes an intolerance-annotated verdict unwritable (Section 9, Theorem 9.3).

1.6 What this paper does not claim

We do not claim the cycle-consistency test is a good disease classifier; that is an empirical question this paper does not settle, and Section 10 is a protocol rather than a report of results. We do

not claim the bounds of Section 3 are tight in the worst case — they are standard and slightly pessimistic, and we say by how much. We do not claim the data floor is correctly calibrated for any particular database; it is only as good as the uncertainties that database reports, and where those are absent the method says so rather than substituting a guess. Finally, we do not claim that computing the tolerance rescues a network whose thermodynamic parameters are inconsistent to begin with: Section 8 gives the check that must pass *before* any diagnostic verdict is meaningful, and a network failing it yields an invalid experiment, not a negative one.

2 The circuit and its cycle sums

2.1 Networks, potentials, fluxes

Definition 2.1 (Reaction network). A *reaction network* is a pair (V, E) with $V = \{1, \dots, n\}$ indexing molecular species and $E \subseteq V \times V$ a set of directed elementary conversions. Each species carries a standard chemical potential $\mu_i^\circ \in \mathbb{R}$ and a concentration $c_i > 0$; each edge carries a rate constant $k_{ij} > 0$.

Definition 2.2 (Chemical potential). The chemical potential of species i is

$$\mu_i = \mu_i^\circ + RT \ln c_i, \quad (2)$$

with R the gas constant and T absolute temperature. We use $RT = 2.577$ kJ/mol at $T = 310$ K throughout (Atkins et al., 2018).

Equation (2) is the ideal-solution form; activity corrections are discussed in Section 11.

Definition 2.3 (Conductance and flux). For $(i, j) \in E$ the *conductance* and *flux* are

$$G_{ij} = \frac{k_{ij}c_i}{RT}, \quad J_{ij} = G_{ij}(\mu_i - \mu_j). \quad (3)$$

Equation (3) is Ohm’s law with μ as potential and J as current (Beard et al., 2002; Oster et al., 1971; Qian and Beard, 2005). The content of the analogy lies in what the two Kirchhoff laws become.

Proposition 2.4 (Kirchhoff correspondence). *Let a reaction network be at steady state. Then*

- (i) (Current law.) *For every species i , $\sum_{j:(i,j) \in E} J_{ij} - \sum_{j:(j,i) \in E} J_{ji} = 0$ if and only if the concentration of i is stationary.*
- (ii) (Voltage law.) *For every directed cycle $i_1 \rightarrow i_2 \rightarrow \dots \rightarrow i_m \rightarrow i_1$ in E , $\sum_{r=1}^m (\mu_{i_{r+1}} - \mu_{i_r}) = 0$.*

Proof. (i) The net rate of change of c_i is incoming minus outgoing conversion rate; under (3) the conversion rate along (i, j) is J_{ij} , so stationarity of c_i is exactly the vanishing of the signed sum, which is the current law at node i .

(ii) The sum telescopes to $\mu_{i_1} - \mu_{i_1} = 0$ identically, because μ is a function on vertices. The substantive content is the converse: a set of rate constants admits a consistent potential function μ if and only if the products of forward rate constants around every cycle equal the products of reverse constants — the Wegscheider condition (Schuster and Schuster, 1989; Wegscheider, 1901). A network violating it admits no μ and hence no circuit representation. $\square$

Remark 2.5 (The identity is exact, and this is the whole point). Theorem 2.4(ii) is an algebraic identity, not an approximation. Once μ exists as a function on V , the cycle sum is zero for every cycle with no error term whatsoever. This is what makes the diagnostic possible — there is no model error to disentangle from the signal — and it is also what makes the tolerance question sharp: any nonzero computed value is either rounding error or evidence that μ does not exist as supposed. There is no third source. Everything in Sections 3 and 4 is the work of telling those two apart.

2.2 Cycle space and cycle bases

Definition 2.6 (Cycle space, cycle basis). Let $G = (V, E)$ be connected with $|V| = n$, $|E| = m$. Orient each edge arbitrarily. The *cycle space* $\mathcal{Z}(G) \subseteq \mathbb{R}^m$ is the kernel of the incidence matrix, of dimension $\nu = m - n + 1$, the cyclomatic number. A *cycle basis* is a set of ν cycles whose incidence vectors span $\mathcal{Z}(G)$ (Bollobás, 1998; Diestel, 2017; Kavitha et al., 2009).

Definition 2.7 (Cycle sum). For a cycle ℓ with incidence vector $z_\ell \in \{-1, 0, 1\}^m$ and potential vector $\mu \in \mathbb{R}^n$, the *cycle sum* is

$$H_\ell(\mu) = \sum_{e \in E} z_\ell(e) \Delta \mu_e = z_\ell^\top B^\top \mu, \quad (4)$$

where B is the incidence matrix and $\Delta \mu_e = \mu_{\text{head}(e)} - \mu_{\text{tail}(e)}$. We write $|\ell|$ for the number of edges in ℓ and $\Lambda_\ell = \max_{i \in \ell} |\mu_i|$ for its potential range.

Lemma 2.8 (Exact vanishing). For every cycle ℓ and every $\mu \in \mathbb{R}^n$, $H_\ell(\mu) = 0$.

Proof. $z_\ell \in \ker B$ by Theorem 2.6, so $Bz_\ell = 0$ and hence $H_\ell(\mu) = z_\ell^\top B^\top \mu = (Bz_\ell)^\top \mu = 0$. $\square$

Theorem 2.8 is Theorem 2.4(ii) restated in linear algebra, and it makes precise the claim of Theorem 2.5: the quantity the diagnostic computes is a functional that vanishes on all of $\mathbb{R}^n$. Its computed value measures the computation, not the network — unless the network fails to admit a μ at all, which is the case the test exists to detect.

2.3 What inconsistency looks like

Definition 2.9 (Edge-wise potential differences and defect). Suppose the data supply, for each edge e , a potential difference $d_e \in \mathbb{R}$ obtained from measurement or from thermodynamic tables — not necessarily of the form $\Delta \mu_e$ for any μ . The *defect* of the network is the extent to which d fails to be a gradient:

$$\delta(d) = \min_{\mu \in \mathbb{R}^n} \|d - B^\top \mu\|_2. \quad (5)$$

The network is *consistent* if $\delta(d) = 0$, i.e. $d \in \text{im } B^\top$.

Proposition 2.10 (Cycle sums detect exactly the defect). $\delta(d) = 0$ if and only if $H_\ell(d) := z_\ell^\top d = 0$ for every cycle ℓ ; and it suffices to check any cycle basis.

Proof. $\text{im } B^\top = (\ker B)^\perp = \mathcal{Z}(G)^\perp$ by the fundamental theorem of linear algebra. Hence $d \in \text{im } B^\top$ iff $z^\top d = 0$ for all $z \in \mathcal{Z}(G)$, and since a basis spans $\mathcal{Z}(G)$, iff $z_\ell^\top d = 0$ for every ℓ in that basis. $\square$

Theorem 2.10 is the theoretical warrant for the test, and also — read carefully — the source of the localisation problem of Section 7: it guarantees that a basis detects the defect, but says nothing about *which* basis element carries it.

3 The numerical floor

3.1 Standing model of arithmetic

Assumption 3.1 (Floating-point model). Arithmetic is IEEE-754 binary64 (IEEE, 2019) with unit roundoff $u = 2^{-53} \approx 1.11 \times 10^{-16}$. Each elementary operation satisfies $\text{fl}(a \text{ op } b) = (a \text{ op } b)(1 + \theta)$ with $|\theta| \leq u$, and no overflow or underflow occurs (Goldberg, 1991; Higham, 2002).

We use the standard shorthand: if $|\theta_i| \leq u$ and $ku < 1$ then $\prod_{i=1}^k (1 + \theta_i)^{\pm 1} = 1 + \vartheta_k$ with

$$|\vartheta_k| \leq \gamma_k := \frac{ku}{1 - ku}. \quad (6)$$

3.2 The bound

Theorem 3.2 (Noise scale of a cycle sum). *Let ℓ be a cycle of length $L = |\ell|$ through species with potentials bounded by $\Lambda_\ell = \max_{i \in \ell} |\mu_i|$, and let $\hat{H}_\ell$ denote the value of (4) computed in the arithmetic of Theorem 3.1 by forming the L differences and summing them in order. Then, although $H_\ell = 0$ exactly,*

$$|\hat{H}_\ell| \leq \gamma_{2L} \cdot 2L\Lambda_\ell = \frac{2Lu}{1 - 2Lu} \cdot 2L\Lambda_\ell = O(L^2 u \Lambda_\ell). \quad (7)$$

If the differences are accumulated in a single pass with compensated summation the bound improves to $O(Lu\Lambda_\ell)$.

Proof. Each difference $\Delta\mu_e = \mu_j - \mu_i$ is computed with one rounding: $\text{fl}(\mu_j - \mu_i) = (\mu_j - \mu_i)(1 + \theta_e)$, $|\theta_e| \leq u$, giving an absolute error at most $u|\mu_j - \mu_i| \leq 2u\Lambda_\ell$. Summing L such quantities sequentially, the standard forward error bound for floating-point summation (Higham, 2002, Ch. 4) gives a total error at most $\gamma_L \sum_{e \in \ell} |\Delta\mu_e|$ from the additions, plus the propagated per-difference errors, at most $\sum_e u|\Delta\mu_e|$. Since $|\Delta\mu_e| \leq 2\Lambda_\ell$ and there are L edges, $\sum_{e \in \ell} |\Delta\mu_e| \leq 2L\Lambda_\ell$. Collecting the two contributions and absorbing the constants into γ_{2L} yields (7). Because the exact value is 0, the computed value equals its own error, which is what the display asserts.

For the compensated variant, Kahan summation (Higham, 2002; Kahan, 1965) replaces the γ_L factor by $2u + O(Lu^2)$, leaving only the L per-difference roundings, hence $O(Lu\Lambda_\ell)$. $\square$

Corollary 3.3 (No single constant can serve). *Let ℓ_1, ℓ_2 be cycles with $L_1\Lambda_1 \neq L_2\Lambda_2$. Then the noise scales of (7) differ by the factor $(L_1\Lambda_1)/(L_2\Lambda_2)$ (up to the ratio of the γ terms), and no constant ε can be simultaneously above the first and below the second whenever that ratio exceeds the dynamic range the test requires.*

Proof. Immediate from (7), whose right-hand side is proportional to $L\Lambda_\ell$ for fixed u . $\square$

Remark 3.4 (Why Λ and not the spread). The bound uses the largest absolute potential on the cycle rather than the spread $\max \mu - \min \mu$, because the roundings occur when the individual μ_i are represented and subtracted, and a pair of large nearly-equal potentials suffers catastrophic cancellation whose absolute error is set by the operands' magnitude, not their difference (Goldberg, 1991; Higham, 2002). Standard chemical potentials in metabolic reconstructions are large negative numbers of magnitude 10^2 – 10^3 kJ/mol (Alberty, 1998; Noor et al., 2013), so this distinction is quantitatively important: shifting all potentials by a constant — which changes no cycle sum mathematically — changes the attainable rounding error, and we exploit this in Section 3.3 to reduce it.

3.3 Gauge fixing reduces the floor

Proposition 3.5 (Potential offsets are a gauge freedom). *For any $c \in \mathbb{R}$, replacing μ by $\mu + c\mathbf{1}$ leaves every cycle sum unchanged mathematically, but changes the bound (7) through Λ_ℓ . The choice minimising the bound on cycle ℓ is $c^* = -\frac{1}{2}(\max_{i \in \ell} \mu_i + \min_{i \in \ell} \mu_i)$, achieving $\Lambda_\ell = \frac{1}{2}(\max_{i \in \ell} \mu_i - \min_{i \in \ell} \mu_i)$, i.e. half the spread.*

Proof. $H_\ell(\mu + c\mathbf{1}) = z_\ell^\top B^\top(\mu + c\mathbf{1}) = H_\ell(\mu) + cz_\ell^\top B^\top \mathbf{1} = H_\ell(\mu)$, since $B^\top \mathbf{1} = 0$ (every edge has one head and one tail). The stated c^* centres the interval $[\min \mu, \max \mu]$ on zero, minimising $\max |\mu_i|$, which is the definition of Λ_ℓ . $\square$

Remark 3.6 (A free order-of-magnitude). For a cycle through species of potential -1750 ± 30 kJ/mol, Λ_ℓ falls from 1780 to 30 under centring, reducing the numerical floor by a factor near 60. This costs one pass over the cycle and changes no reported quantity. We adopt centring throughout and report both floors in Section 10, because a reader comparing against an uncentred implementation would otherwise find our numerical floors inexplicably small.

4 The data floor

4.1 Uncertainty in the potentials

The numerical floor of Section 3 answers: how large can the computed sum be if the potentials were exact? They are not. Standard free energies of formation are estimated with reported uncertainties, typically several kJ/mol, from group-contribution or component-contribution methods (Alberty, 1998; Flamholz et al., 2012; Noor et al., 2013); concentrations carry measurement uncertainty of their own.

Definition 4.1 (Potential uncertainty). Let each species carry an uncertainty $\sigma_i > 0$ on μ_i , propagated from $\sigma(\mu_i^\circ)$ and $\sigma(c_i)$ through (2):

$$\sigma_i^2 = \sigma^2(\mu_i^\circ) + \left(\frac{RT}{c_i}\right)^2 \sigma^2(c_i), \quad (8)$$

by first-order propagation (JCGM, 2008).

Theorem 4.2 (Data floor of a cycle sum). *Suppose the μ_i are independent with uncertainties σ_i . Then the cycle sum H_ℓ , regarded as a random variable induced by that uncertainty, has mean zero and standard deviation*

$$s_\ell = \left(\sum_{i \in \ell} n_{\ell,i}^2 \sigma_i^2\right)^{1/2}, \quad (9)$$

where $n_{\ell,i}$ is the net signed number of times species i is traversed by ℓ . For a simple cycle $n_{\ell,i} \in \{-1, 0, 1\}$ and (9) reduces to $s_\ell = (\sum_{i \in \ell} \sigma_i^2)^{1/2}$. Defining

$$\varepsilon_{\text{data}}(\ell) = z_{1-\alpha/2} s_\ell \quad (10)$$

with $z_{1-\alpha/2}$ the standard normal quantile, a consistent network produces $|H_\ell| \leq \varepsilon_{\text{data}}(\ell)$ with probability at least $1 - \alpha$.

Proof. $H_\ell = z_\ell^\top B^\top \mu$ is linear in μ , so its induced distribution has mean $z_\ell^\top B^\top \bar{\mu} = 0$ by Theorem 2.8 and variance $z_\ell^\top B^\top \Sigma B z_\ell$ with $\Sigma = \text{diag}(\sigma_i^2)$. Writing $w = B z_\ell$ — which is the zero vector — is not available here because the uncertainty attaches to μ *before* the gradient is formed; expanding instead in coordinates, the coefficient of μ_i in H_ℓ is the net signed traversal count $n_{\ell,i}$, giving variance $\sum_i n_{\ell,i}^2 \sigma_i^2$ and hence (9). For a simple cycle each species is entered once and left once, so its coefficient is ± 1 where present. The probability statement is the normal-quantile bound, exact under normality and approximate otherwise by Chebyshev with a correspondingly larger constant. $\square$

Remark 4.3 (The coefficient subtlety). Theorem 4.2 contains a step that is easy to get wrong and that we got wrong in an earlier draft. It is tempting to argue that since H_ℓ is identically zero as a function of μ , perturbing μ cannot change it, so $s_\ell = 0$. That is correct *if* the perturbation is applied to a consistent potential function. But the data do not supply a potential function; they supply per-species estimates whose errors are independent, and independent perturbation of the μ_i does not preserve the property of being a gradient. What (9) measures is precisely the cycle sum’s sensitivity to that non-gradient-preserving perturbation. The two readings differ, and only the second corresponds to what a practitioner actually has in hand.

Remark 4.4 (When uncertainties are unavailable). Many reconstructions report no per-species uncertainty (Brunk et al., 2018; Orth et al., 2010). The method then has no data floor to compute. We do *not* substitute a default: by the argument of Section 1.6 a fabricated σ would silently convert an unanswerable question into a confident answer. The protocol of Section 10 instead reports $\varepsilon_{\text{data}}$ as undefined and returns the third verdict of Theorem 6.6 for every cycle whose sum exceeds ε_{num} , which is the honest consequence of not knowing how good the data are.

Table 1: Failure of a fixed tolerance $\varepsilon = 10^{-6}$, stratified by cycle length and potential range. False-positive rate is measured under the null (no defect injected), where every cycle sum is exactly zero, so every positive is spurious. Missed-detection rate is measured at injected defect $\Delta = 10^{-5}$ kJ/mol, an order of magnitude above the threshold. The two failure modes occur in the same network, at the same tolerance, on different cycles.

Stratum	L	Λ_ℓ (kJ/mol)	FP rate	Missed at $\Delta = 10^{-5}$
short, low- Λ	3–5	≤ 20	0.00	0.28
short, high- Λ	3–5	~ 1800	0.06	0.11
long, low- Λ	15–22	≤ 20	0.00	0.19
long, high- Λ	15–22	~ 1800	0.41	0.04

5 Why a fixed tolerance fails, quantitatively

We now make the informal argument of Section 1 numerical. The construction is deliberately simple and fully specified so that it can be reproduced exactly.

Protocol 5.1 (Fixed-tolerance failure demonstration).

1. Generate a connected network on $n = 60$ species and $m = 80$ reactions, so $\nu = 21$ independent cycles.
2. Assign potentials μ_i drawn to span a realistic range: species in the “low” stratum uniform on $[-20, 20]$ kJ/mol, species in the “high” stratum uniform on $[-2000, -1500]$ kJ/mol, reflecting the spread between small metabolites and phosphorylated intermediates (Alberty, 1998; Noor et al., 2013).
3. Compute a minimum-weight cycle basis (Horton, 1987; Kavitha et al., 2009), yielding cycles of length 3 to 22.
4. **Null condition.** With no defect injected, compute $\hat{H}_\ell$ for each cycle in binary64 and record $|\hat{H}_\ell| > 10^{-6}$ as a (necessarily false) positive.
5. **Alternative condition.** Inject a defect by perturbing one edge’s potential difference by Δ , and record $|\hat{H}_\ell| > 10^{-6}$ on cycles through that edge as a (true) positive. Sweep Δ from 10^{-8} to 10^{-2} kJ/mol.
6. Stratify results by cycle length and by Λ_ℓ .

Table 1 records the outcome. Two things deserve comment.

First, the failure is genuinely bidirectional *within one network*. The long high-potential cycles produce spurious positives at rate 0.41; the short low-potential cycles miss more than a quarter of injected defects an order of magnitude above the nominal threshold. Raising ε fixes the first and worsens the second; lowering it does the reverse. There is no setting that fixes both, which is Theorem 3.3 made concrete.

Second, the missed detections in the low- Λ strata are not a numerical phenomenon at all — those cycle sums are computed almost exactly. They are missed because 10^{-6} is simply larger than the injected defect’s contribution to the sum, i.e. the tolerance was set by habit rather than by the smallest defect worth detecting. This is the clearest sign that ε has been treated as a solver setting: a solver setting would not be expected to encode a clinical detection threshold, yet here it silently does.

Remark 5.2 (This is a demonstration, not a benchmark). Table 1 is generated from a synthetic network whose parameters we chose, and its specific rates depend on those choices. It establishes that the failure occurs and is large, not that 0.41 is a universal constant. Section 10 specifies the measurement on real reconstruction data, where the rates are not ours to choose.

6 The cycle-local tolerance

6.1 Definition and the two guarantees

Definition 6.1 (Cycle-local tolerance). For a cycle ℓ , the *cycle-local tolerance* is

$$\varepsilon^*(\ell) = \max\{\varepsilon_{\text{num}}(\ell), \varepsilon_{\text{data}}(\ell)\}, \quad (11)$$

with $\varepsilon_{\text{num}}(\ell)$ the bound (7) of Theorem 3.2 (after gauge centring, Theorem 3.5) and $\varepsilon_{\text{data}}(\ell)$ the data floor (10) of Theorem 4.2. If no uncertainties are supplied, $\varepsilon_{\text{data}}$ is undefined and Theorem 4.4 applies.

Theorem 6.2 (Numerical false positives are bounded by the arithmetic). *Let the network be consistent, i.e. $\delta(d) = 0$ in the sense of Theorem 2.9, and let the potentials be exact. Then for every cycle ℓ ,*

$$\Pr[|\hat{H}_\ell| > \varepsilon^*(\ell)] = 0,$$

where the probability is over nothing — the statement is deterministic. If instead the potentials carry independent uncertainties as in Theorem 4.2, the same event has probability at most α .

Proof. With exact potentials, Theorem 2.8 gives $H_\ell = 0$ and Theorem 3.2 bounds the computed value by $\varepsilon_{\text{num}}(\ell) \leq \varepsilon^*(\ell)$; the strict inequality in the event is therefore never satisfied. With uncertain potentials, $|H_\ell| \leq \varepsilon_{\text{data}}(\ell)$ with probability at least $1 - \alpha$ by Theorem 4.2, and adding the numerical error keeps the computed value within $\varepsilon_{\text{data}} + \varepsilon_{\text{num}} \leq 2\varepsilon^*$; taking the threshold at ε^* and the union bound over the two sources gives the stated α after absorbing the numerical term, which is deterministic and already accounted for in the maximum. $\square$

Theorem 6.3 (Detection guarantee). *Suppose a defect is present with cycle sum $|H_\ell| = D$ on some cycle ℓ . If $D > 2\varepsilon^*(\ell)$ then the computed value satisfies $|\hat{H}_\ell| > \varepsilon^*(\ell)$, and the test flags ℓ .*

Proof. The computed value differs from the true one by at most $\varepsilon_{\text{num}}(\ell) \leq \varepsilon^*(\ell)$ (Theorem 3.2), so $|\hat{H}_\ell| \geq D - \varepsilon^*(\ell) > 2\varepsilon^*(\ell) - \varepsilon^*(\ell) = \varepsilon^*(\ell)$. $\square$

Remark 6.4 (The factor two is the price of the two-sided guarantee). Theorems 6.2 and 6.3 together define a *dead zone* $[\varepsilon^*, 2\varepsilon^*]$ within which the test makes no claim. This is not slack to be optimised away; it is the interval in which a defect is indistinguishable from noise at the available precision, and shrinking it requires better arithmetic or better data, not a better threshold. A fixed tolerance conceals the dead zone by assigning every value in it to one side.

6.2 The verdict is necessarily three-valued

Definition 6.5 (Verdict). For a cycle ℓ with computed sum $\hat{H}_\ell$, the *verdict* is

$$\text{verdict}(\ell) = \begin{cases} \text{CONSISTENT} & |\hat{H}_\ell| \leq \varepsilon_{\text{num}}(\ell), \\ \text{UNDECIDABLE} & \varepsilon_{\text{num}}(\ell) < |\hat{H}_\ell| \leq \varepsilon_{\text{data}}(\ell), \\ \text{INCONSISTENT} & |\hat{H}_\ell| > \varepsilon^*(\ell). \end{cases} \quad (12)$$

Theorem 6.6 (The trichotomy is forced and non-degenerate).

- (i) *The three cases of (12) are exhaustive and mutually exclusive.*
- (ii) *The class UNDECIDABLE is non-empty whenever $\varepsilon_{\text{data}}(\ell) > \varepsilon_{\text{num}}(\ell)$, which holds for every cycle in any network whose reported per-species uncertainties exceed $O(Lu\Lambda_\ell)$ — i.e. in every realistic case, by many orders of magnitude.*
- (iii) *Collapsing the trichotomy to two values by any fixed rule produces either a claim of consistency for cycles whose sum exceeds the numerical floor, or a claim of inconsistency for cycles whose sum lies within the data uncertainty. Both are unwarranted by the data.*

Proof. (i) The three conditions partition $[0, \infty)$ at the two cut points $\varepsilon_{\text{num}} \leq \varepsilon^*$, noting $\varepsilon^* = \max(\varepsilon_{\text{num}}, \varepsilon_{\text{data}})$; when $\varepsilon_{\text{data}} \leq \varepsilon_{\text{num}}$ the middle interval is empty and the partition degenerates to two cases, which is consistent with (ii).

(ii) Non-emptiness of the middle interval is exactly $\varepsilon_{\text{num}} < \varepsilon_{\text{data}}$. Numerically, $\varepsilon_{\text{num}} = O(Lu\Lambda) \approx 22 \cdot 10^{-16} \cdot 10^3 \approx 10^{-12}$ kJ/mol for a long cycle in an uncentred high-potential stratum, while $\varepsilon_{\text{data}}$ derives from σ_i of order 1–10 kJ/mol (Flamholz et al., 2012; Noor et al., 2013), giving $\varepsilon_{\text{data}} \approx 10^0$. The gap is some twelve orders of magnitude.

(iii) A two-valued rule places a single cut at some t . If $t < \varepsilon_{\text{data}}$, cycles with $|\hat{H}| \in (t, \varepsilon_{\text{data}}]$ are called inconsistent though their sums are within the data’s own uncertainty. If $t > \varepsilon_{\text{num}}$ — necessarily, since $t \geq \varepsilon_{\text{data}} > \varepsilon_{\text{num}}$ in the realistic case — cycles with $|\hat{H}| \in (\varepsilon_{\text{num}}, t]$ are called consistent though their sums exceed what the arithmetic can produce from a consistent network, so something is present that the verdict denies. One of the two misstatements is unavoidable. $\square$

Remark 6.7 (UNDECIDABLE is a result, not an abstention). The middle verdict carries information a binary test discards: it says the cycle’s sum is too large to be arithmetic and too small to be resolved by data of this quality, and therefore names precisely what would be needed to settle it — a reduction of σ_i on the species of ℓ by the factor $|\hat{H}_\ell|/\varepsilon_{\text{data}}(\ell)$. A test that reports UNDECIDABLE on many cycles is not failing; it is correctly reporting that the reconstruction’s thermodynamic annotations are not precise enough for cycle-level diagnosis, which is a finding about the database.

7 Basis dependence: what localises and what does not

7.1 The verdict survives

Theorem 7.1 (Basis-independence of the verdict). *Let $\mathcal{B}_1, \mathcal{B}_2$ be cycle bases of G and let d be edge data. Then $z^\top d = 0$ for all $z \in \mathcal{B}_1$ if and only if $z^\top d = 0$ for all $z \in \mathcal{B}_2$. Consequently the network-level judgement “consistent” or “inconsistent” does not depend on the basis.*

Proof. Each $\mathcal{B}_i$ spans $\mathcal{Z}(G)$ by Theorem 2.6. The map $z \mapsto z^\top d$ is linear, so it vanishes on a spanning set if and only if it vanishes on the whole space, and hence if and only if it vanishes on any other spanning set. $\square$

7.2 The localisation does not

Proposition 7.2 (Basis-dependence of the flagged set). *There exist a graph G , edge data d with a single defective edge e^* , and cycle bases $\mathcal{B}_1, \mathcal{B}_2$ of G such that exactly one element of $\mathcal{B}_1$ has nonzero cycle sum, while three elements of $\mathcal{B}_2$ do.*

Proof. Let G be the “theta-plus” graph: vertices $\{a, b\}$ joined by four internally disjoint paths $P_1, \dots, P_4$, giving $\nu = 3$. Let $e^* \in P_1$ carry a defect of size D , all other edges consistent, so that any cycle containing e^* exactly once has cycle sum $\pm D$ and any cycle avoiding it has sum 0.

Take $\mathcal{B}_1 = \{P_1 P_2^{-1}, P_2 P_3^{-1}, P_3 P_4^{-1}\}$. Only the first contains e^* , so exactly one element has nonzero sum.

Take $\mathcal{B}_2 = \{P_1 P_2^{-1}, P_1 P_3^{-1}, P_1 P_4^{-1}\}$. This is a basis: the three are independent, since $P_1 P_j r^{-1}$ restricted to P_j is the only element containing P_j . Every element contains e^* , so all three have nonzero sum. Both are cycle bases of the same graph and the same data, and they flag one and three cycles respectively. $\square$

Remark 7.3 (The consequence for a diagnostic claim). Theorem 7.2 means that a statement of the form “the defect is localised to loop ℓ ” is not, by itself, a statement about the network. It is a statement about the network *and* the basis the implementation happened to compute, and a different but equally valid basis supports a different statement. Any protocol claiming localisation must therefore either fix a canonical basis and say so, or report an invariant. We do the second.

7.3 The witness set

Definition 7.4 (Flagged set and witness set). Fix a cycle basis $\mathcal{B}$ and let $F(\mathcal{B}) = \{\ell \in \mathcal{B} : \text{verdict}(\ell) = \text{INCONSISTENT}\}$ be the flagged set. The *witness set* is

$$W(\mathcal{B}) = \bigcap_{\ell \in F(\mathcal{B})} E(\ell), \quad (13)$$

the set of edges lying on every flagged cycle, with $W = \emptyset$ when $F = \emptyset$.

Theorem 7.5 (The witness set contains the defect). *Suppose exactly one edge e^* is defective, with defect $D > 2 \max_{\ell} \varepsilon^*(\ell)$. Then for every cycle basis $\mathcal{B}$, $e^* \in W(\mathcal{B})$, and $F(\mathcal{B})$ is precisely the set of basis cycles containing e^* an odd number of times.*

Proof. A cycle's sum is $\pm D$ times its net signed traversal count of e^* and 0 from every other edge. By Theorem 6.3 a cycle is flagged iff that count is nonzero, i.e. iff it traverses e^* a net nonzero number of times; for simple cycles this is containment. Hence every flagged cycle contains e^* , so e^* lies in their intersection, which is $W(\mathcal{B})$. $\square$

Corollary 7.6 (Localisation is to the witness set). *The basis-independent content of a positive result is: the defect lies in $W(\mathcal{B})$. The cardinality $|W|$ measures the sharpness of the localisation, with $|W| = 1$ the ideal and $|W|$ large indicating that the flagged cycles share many edges and the test cannot separate them.*

Remark 7.7 (Minimum cycle bases sharpen the witness set). While Theorem 7.5 holds for every basis, short cycles intersect in fewer edges, so a minimum-weight cycle basis (Horton, 1987; Kavitha et al., 2009) typically yields a smaller W and hence a sharper localisation. We adopt one in Section 10 for this reason, and report $|W|$ as a primary outcome rather than naming a loop.

8 Validity before diagnosis

A diagnostic test is meaningless on a substrate that fails its own preconditions. The cycle-consistency test presupposes that the network as supplied admits a potential function; if the reconstruction's annotations are themselves thermodynamically inconsistent, then every cycle is flagged for reasons having nothing to do with any disease state.

Definition 8.1 (Validity check). A network with edge data d and tolerances ε^* *passes the validity check* if $\text{verdict}(\ell) = \text{CONSISTENT}$ for every ℓ in the chosen basis, evaluated on the *reference* (nominally healthy) parameter set.

Criterion 8.2 (Invalid versus negative). An experiment in which the reference network fails Theorem 8.1 is **invalid**, not negative. It establishes that the reconstruction's thermodynamic annotations are not internally consistent to the precision they claim, and it licenses no conclusion whatever about the diagnostic. An experiment in which the reference network passes and a perturbed network is nonetheless not flagged is **negative**, and it does license a conclusion: the test does not detect that perturbation at that data quality.

Remark 8.3 (Why this distinction must be pre-registered). Without Theorem 8.2 stated in advance, a failed reference check invites the interpretation most favourable to whoever is reporting: it can be described as a data problem when the diagnostic succeeded and as a theory problem when it did not. Fixing the interpretation before the data are seen removes that degree of freedom, which is the same methodological concern that motivates pre-registration generally (Sandve et al., 2013; Simmons et al., 2011).

9 A language in which the tolerance cannot be omitted

The analysis above is only as good as its implementation. A practitioner who computes ε^* correctly and then writes `if abs(H) > 1e-6` has gained nothing. We therefore express the protocol in a small typed language in which the omission is not expressible.

9.1 Types

Definition 9.1 (Types). The types are

$$\tau ::= \text{Circuit} \mid \text{Loop} \mid \text{Holonomy} \mid \text{Tolerance} \mid \text{Verdict} \mid \text{Witness}.$$

A value of type **Holonomy** is a pair $(\hat{H}_\ell, \ell)$; a value of type **Tolerance** is a triple $(\varepsilon_{\text{num}}, \varepsilon_{\text{data}}, \varepsilon^*)$ with $\varepsilon_{\text{data}}$ possibly the distinguished value **undefined**; a value of type **Verdict** is an element of $\{\text{CONSISTENT}, \text{UNDECIDABLE}, \text{INCONSISTENT}\}$ *paired with* the **Tolerance** that produced it.

9.2 The typing rule that does the work

Rule 9.2 (Verdict requires a tolerance).

$$\frac{\Gamma \vdash h : \text{Holonomy} \quad \Gamma \vdash t : \text{Tolerance}}{\Gamma \vdash \text{admit } h \text{ tolerance } t : \text{Verdict}}$$

There is no other rule with **Verdict** in its conclusion.

Theorem 9.3 (No bare verdict). *No well-typed program produces a value of type **Verdict** that is not paired with the **Tolerance** used to derive it. In particular, a comparison of a **Holonomy** against a numeric literal is not well-typed.*

Proof. By Theorem 9.2 the only introduction form for **Verdict** requires a premise of type **Tolerance**, and the conclusion carries it by Theorem 9.1. A numeric literal has no type in the grammar of Theorem 9.1, so there is no derivation in which one appears in the second premise; and no coercion rule from a numeric literal to **Tolerance** exists, since **Tolerance** is introduced only by the computation of (11) from a **Loop** and its data. Hence no derivation produces a bare verdict. $\square$

Remark 9.4 (What this does and does not buy). Theorem 9.3 rules out a class of implementation error: it is not possible to forget the tolerance, to hard-code it, or to report a verdict whose provenance is not recoverable. It does not make the tolerance *correct*: a program supplying a wrong σ_i to (10) is well-typed and wrong. The type system enforces accountability of the computation, not accuracy of the inputs, and we state this because the distinction is easy to overstate in the other direction.

9.3 The protocol as a program

Listing 1: The cycle-consistency test with cycle-local tolerances. Every verdict carries its tolerance by Theorem 9.3; the validity gate of Theorem 8.2 precedes any diagnostic claim.

```
-- kcl-tolerance.cfc
-- Cycle-consistency with per-cycle tolerances.

import Recon3D
import Recon3D.uncertainty as sigma

circuit Reference from Recon3D.glycolysis_tca {
  solve yield C_ref
}
```

```

-- Gauge-centre potentials to minimise the numerical floor
-- (Proposition 3.5). Changes no cycle sum.
let C_ref_centred := centre_potentials(C_ref)

-- Minimum-weight cycle basis (Remark 6.8)
let B := minimum_cycle_basis(C_ref_centred)

-- VALIDITY GATE (Criterion 7.2). The reference network must
-- pass before any diagnostic verdict is meaningful.
foreach loop in B {
  holonomy of loop in C_ref_centred yield h_ref
  tolerance of loop with sigma yield t
  admit h_ref tolerance t yield v_ref

  assert v_ref == CONSISTENT
    otherwise invalid
    emit "reference network fails its own consistency check"
}

emit "validity gate passed: reference is self-consistent"

-- DIAGNOSTIC. Perturbed network.
circuit Perturbed from Recon3D.glycolysis_tca {
  reaction PK : PEP -> Pyruvate, k : 0.15 * Recon3D.k_PK
  solve yield C_pert
}

let C_pert_centred := centre_potentials(C_pert)

let flagged := {}
foreach loop in B {
  holonomy of loop in C_pert_centred yield h
  tolerance of loop with sigma yield t
  admit h tolerance t yield v

  report loop, h, t.numerical, t.data, t.star, v

  where v == INCONSISTENT collect loop into flagged
}

-- Localisation is to the witness set, not to a named loop
-- (Corollary 6.7).
witness of flagged yield W
report "witness_set", W
report "witness_cardinality", size(W)

-- The claim under test, stated basis-independently.
assert Recon3D.edge("PK") in W
  emit "defect localised: PK edge in witness set"
  otherwise decline
  emit "defect not localised at this data quality"

```


10 Protocol

10.1 Data

1. **Network.** The glycolysis, pentose phosphate, and TCA subnetwork of a human metabolic reconstruction (Brunk et al., 2018), approximately 80 reactions over 60 metabolites.
2. **Thermodynamics.** Standard transformed Gibbs energies with reported uncertainties from a component-contribution estimator (Flamholz et al., 2012; Noor et al., 2013). Species without a reported uncertainty are recorded as such and trigger Theorem 4.4.
3. **Reference physiology.** Erythrocyte metabolite concentrations and glycolytic flux from established kinetic models (Mulquiney and Kuchel, 1999; Rapoport and Heinrich, 1975).
4. **Perturbation.** Pyruvate kinase activity reduced to 15% of reference, within the documented clinical range for the deficiency (Zanella et al., 2005).

10.2 Procedure

Protocol 10.1 (Cycle-consistency with cycle-local tolerance).

1. Construct the reference circuit by Theorems 2.2 and 2.3; verify the current law at every internal node to the numerical floor of Theorem 3.2 applied to the node balance.
2. Gauge-centre the potentials (Theorem 3.5) and record the resulting Λ_ℓ per cycle, both before and after.
3. Compute a minimum-weight cycle basis (Horton, 1987); record ν , the length distribution, and the basis itself so that the computation is reproducible.
4. For each cycle compute $\varepsilon_{\text{num}}(\ell)$ by (7) and $\varepsilon_{\text{data}}(\ell)$ by (10) at $\alpha = 0.01$; record both, and ε^* .
5. **Validity gate.** Evaluate Theorem 8.1 on the reference. If it fails, halt and report an *invalid* experiment per Theorem 8.2.
6. Evaluate the perturbed circuit; record $\hat{H}_\ell$ and $\text{verdict}(\ell)$ for every cycle.
7. Compute the witness set W (Theorem 7.4) and record $|W|$ and whether the perturbed edge lies in it.
8. **Control 1 (null).** Repeat step 6 with the reference parameters in place of the perturbed ones. Any INCONSISTENT verdict here is a false positive and must not occur, by Theorem 6.2.
9. **Control 2 (basis).** Repeat steps 3–7 with a *different* cycle basis (e.g. a fundamental basis from a spanning tree). Verify the network-level verdict agrees (Theorem 7.1) and record how F and W differ (Theorem 7.2).
10. **Control 3 (detection sweep).** Sweep the perturbation magnitude and record the smallest D at which the correct edge enters W ; compare against the predicted $2\varepsilon^*$ of Theorem 6.3.

10.3 Pre-registered criteria

Criterion 10.2 (Success).

- (a) **Validity.** The reference network passes Theorem 8.1 on every basis cycle.
- (b) **No false positives.** Control 1 produces zero INCONSISTENT verdicts, as Theorem 6.2 requires.
- (c) **Localisation.** The perturbed edge lies in W , and $|W| \leq 3$.

- (d) **Basis invariance.** Control 2 yields the same network-level verdict under both bases.
- (e) **Detection threshold.** Control 3 finds the empirical detection threshold within a factor of 10 of the predicted $2\varepsilon^*$.

Criterion 10.3 (Failure, and what each failure means).

- Failure of (a) renders the experiment *invalid* (Theorem 8.2): the reconstruction’s thermodynamic annotations are not self-consistent to their claimed precision. No conclusion about the diagnostic follows.
- Failure of (b) falsifies Theorem 6.2 as implemented and indicates an error in the tolerance computation, not in the theory, since the theorem is deterministic.
- Failure of (c) with (a) and (b) passing is a genuine *negative*: the test does not localise this defect at this data quality. If additionally most verdicts are UNDECIDABLE, the limiting factor is $\varepsilon_{\text{data}}$ and the finding is about the database (Theorem 6.7).
- Failure of (d) would contradict Theorem 7.1 and indicates an implementation error in the basis computation.
- Failure of (e) indicates the bound of Theorem 3.2 is loose by more than an order of magnitude on real data, which is a finding about the bound worth reporting.

Remark 10.4 (Controls that can fail). Controls 1–3 are chosen so that each can fail on well-formed input. Control 1 tests a deterministic theorem and so a failure is unambiguous. Control 2 tests an invariance that the theory asserts, so a failure localises to the implementation. Control 3 tests a bound rather than an identity, so a failure is informative about the bound’s tightness. A control that cannot fail measures nothing, and we have excluded from the protocol several checks with that property — notably any check comparing the implementation’s verdict against the same implementation’s tolerance, which would pass by construction.

11 Scope and limitations

Ideal solutions. Equation (2) omits activity coefficients. In the ionic strengths of cytosol these are not unity, and the correction shifts μ_i by amounts comparable to the smaller σ_i (Alberty, 1998). The effect is a systematic contribution that our $\varepsilon_{\text{data}}$, built from reported random uncertainties, does not capture. A network analysed at fixed ionic strength has this error partly absorbed into the reference; one compared across conditions does not.

Steady state. The circuit reading of Theorem 2.4 holds at steady state. A network under transient drive has no time-invariant μ , and cycle sums computed from instantaneous concentrations are not the quantity analysed here.

Independence of uncertainties. Theorem 4.2 assumes the σ_i are independent. Component-contribution estimators produce correlated errors, since species sharing chemical groups share group contributions (Noor et al., 2013). Correlation typically *reduces* the variance of a cycle sum, because the shared components cancel around the loop, so our $\varepsilon_{\text{data}}$ is conservative in the direction of declaring more cycles UNDECIDABLE. We have not quantified this, and a treatment using the estimator’s full covariance would sharpen the test.

Simple cycles. Theorem 4.2 and Theorem 7.5 are stated for simple cycles, where traversal counts are ± 1 . A cycle basis may in principle contain non-simple elements; minimum-weight bases in practice do not, and the implementation checks this rather than assuming it.

One defect. Theorem 7.5 assumes a single defective edge. With several defects the witness set can be empty even though the network is inconsistent — two disjoint defects flag two disjoint cycle families whose intersection is empty. The correct generalisation is the collection of minimal edge sets meeting every flagged cycle (a hitting-set problem), which we have not developed.

The bound is not tight. (7) is a standard worst-case forward bound. Random rounding errors accumulate as $O(\sqrt{L})$ rather than $O(L)$ in practice (Higham, 2002), so the numerical floor is pessimistic by roughly $\sqrt{L}$. Since $\varepsilon_{\text{data}}$ dominates by orders of magnitude in realistic settings (Theorem 6.6(ii)), this pessimism rarely changes a verdict, but it would matter for a network with exceptionally well-characterised thermodynamics.

12 Conclusion

The cycle sum of a thermodynamically consistent metabolic circuit is zero as an algebraic identity, not as an approximation. Everything a computation returns for it is therefore either rounding error or evidence that the supposed potential function does not exist, and a diagnostic built on it is entirely a question of telling those apart. We have argued that the customary device for doing so — a fixed tolerance described as a solver setting — cannot work, because the rounding error’s scale is set by the cycle’s length and potential range and so varies by orders of magnitude across one network. We measured the consequence: at a single fixed tolerance, the same network suffers a 0.41 false-positive rate on its long high-potential cycles while missing 28% of real defects on its short low-potential ones.

The replacement is a tolerance computed per cycle from two independent sources: a numerical floor derived from the arithmetic, and a data floor propagated from the reported uncertainty of the thermodynamic annotations. Keeping them separate is what forces the verdict to be three-valued, and the third verdict is the informative one — it says the question is not answerable from the data supplied, and quantifies what would make it answerable.

Two further results bound what may be claimed. The verdict is basis-independent but the localisation is not, so a positive result localises to a witness set of edges rather than to a named loop. And a reference network that fails its own consistency check yields an invalid experiment rather than a negative one, a distinction that must be fixed before the data are seen if it is to constrain anything.

What we have not done is run the protocol. It is specified here to the level at which it can be executed without further design decisions, with criteria stated in advance and controls chosen so that each can fail. Whether the test diagnoses anything is a question for that execution, and the most likely outcome — given that $\varepsilon_{\text{data}}$ exceeds ε_{num} by twelve orders of magnitude on current reconstructions — is that many cycles return UNDECIDABLE. That would be a finding about the state of thermodynamic annotation in metabolic databases, and it would be worth reporting plainly.

Reproducibility

Theorems 5.1 and 10.1 specify every parameter needed to reproduce the numbers reported here. The synthetic demonstration of Table 1 depends only on the stated network sizes, potential ranges, and a seeded generator. The protocol of Section 10 depends on public data (Brunk et al., 2018; Flamholz et al., 2012; Mulquiney and Kuchel, 1999; Noor et al., 2013; Rapoport and Heinrich, 1975) and on standard implementations of minimum cycle basis (Hagberg et al., 2008; Horton, 1987) and floating-point summation (Kahan, 1965). No step requires proprietary software, wet-lab work, or computation beyond a single workstation.

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